Anshul Raje

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US Citizen | Seeking Summer 2027 Internship | Robotics Perception & Computer Vision

EDUCATION

M.Sc. in Robotics Engineering, Worcester Polytechnic Institute, Worcester, MA — **Aug 2026 – May 2028**
Courses: Vision-based Robotic Manipulation, Hands-on Autonomous Aerial Robotics

B.E. in Electronics and Communication Engineering, BITS Pilani, Goa, India — **Sep 2020 – Jul 2024**
GPA: 3.40/4.0 | Courses: Digital Image Processing, Control Systems, Digital Signal Processing

SKILLS

Languages:	Python, C/C++
Frameworks & Libraries:	TensorFlow, PyTorch, ROS 2, ROS 2 Control, OpenCV, PCL, Ultralytics
Tools & Platforms:	Linux, Git, Docker, Gazebo (robot simulation), RViz (visualization)

EXPERIENCE

Strider Robotics | Associate Robotics Engineer | Bangalore, India — **Aug 2024 – Apr 2026**

- Automated analog gauge reading by detecting needle base, needle tip, and the min and max scale markers, then converting the angle between them into a calibrated value, reading 22 of 25 gauges correctly per run on site.
- Built a vision-guided capture pipeline servoing a PTZ camera onto a known ArUco marker and then onto the gauge itself, zooming and autofocusing to image each asset from a repeatable viewpoint every round.
- Cut inference cost in the alignment loop by running YOLO with DeepSORT tracking only on the first pass, then switching to normalized cross-correlation template matching for every later alignment step.
- Trained YOLOv12 detectors on 2500+ hand-labelled images spanning gauge, needle, scale marker, and fire extinguisher classes, converting them to TensorRT engines to run 720p inference at 25 FPS onboard a Jetson.
- Designed a five-layer ROS 2 perception architecture separating camera drivers, computer vision inference nodes, and task orchestration, and deployed it on legged inspection robots at two industrial clients.
- Architected a behavior tree orchestration framework separating reusable perception primitives from client-specific task sequences, letting the team add a new client's inspection routine as a configuration package in one to two hours.
- ***Tools & Techniques:*** *Visual Servoing, ArUco Marker Detection, Multi-Object Tracking, Template Matching, TensorRT, Edge Inference, Behavior Trees, DeepSORT.*

HiPeRT Lab, UNIMORE | Robotics Research Intern | Modena, Italy — **Jul 2023 – Dec 2023**

- Engineered a synthetic dataset pipeline segmenting track footage with Segment Anything Model (SAM) and compositing opponent car CAD models per frame, auto-annotating 15000+ images with zero manual labeling.
- Trained YOLOv4 on domain-randomized frames varying lighting, saturation, and blur, hitting a 97% detection rate on live on-track video and adding camera perception to a LiDAR-only race car stack.
- ***Tools & Techniques:*** *Synthetic Data Generation, Domain Randomization, Real-Time Inference, SAM, YOLOv4.*

PROJECTS

Quadrotor Navigation with RRT* Planning and Quintic Trajectories [Report] [Video] — **Sep 2026 – Oct 2026**

- Implemented an RRT* planner with goal-biased sampling, lowest-cost parent selection, and rewiring, then simplified its paths by greedy shortcutting against inflated obstacles to navigate a quadrotor through known 3D maps.
- Generated quintic spline trajectories continuous through snap by solving a block tridiagonal system for waypoint velocities and accelerations, scaling segment times to hold velocity and acceleration limits.
- Tuned the position and velocity loops of a cascaded PID controller to a mean tracking error of 14 cm in simulation, reaching the goal without collision in 500+ randomized runs across two maps.
- Flew a Crazyflie through a course of physical boxes under Vicon motion capture with a pure pursuit policy sending feedforward velocity commands, reaching the goal in 5.8 s with a 2.9 cm mean path error and no collision.
- ***Tools & Techniques:*** *Motion Planning, RRT*, Collision Checking, Trajectory Generation, Quintic Splines, Cascaded PID Control, Pure Pursuit, Quadrotor Simulation, Vicon Motion Capture, Sim-to-Real Transfer.*

Attitude Estimation using a 6-DoF IMU [Report] [Video] — **Aug 2026 – Sep 2026**

- Benchmarked five attitude estimation methods against Vicon motion capture ground truth across six flight sequences, with an unscented Kalman filter reaching the lowest mean orientation RMSE of 14.6 degrees.
- Implemented a quaternion-based unscented Kalman filter fusing accelerometer data with a bias-calibrated gyroscope, propagating sigma points through the nonlinear process model to hold roll and pitch error to 5.4 and 4.8 degrees.
- ***Tools & Techniques:*** *State Estimation, Sensor Fusion, Unscented Kalman Filter (UKF), Madgwick Filter, Complementary Filter, Quaternion Attitude Representation, IMU Calibration, RMSE Benchmarking.*

Autonomous Navigation and Manipulation for a Mars Rover [Code] [Video] **Sep 2021 – May 2023**

- Added real-time obstacle detection to an autonomous rover by training YOLOv3 on a custom dataset, letting the rover identify and avoid hazards on course without operator input.
- Implemented a Stanley controller for autonomous waypoint navigation, holding tracking error within 5 cm across the traversal course.
- Developed forward and inverse kinematics for two joints of a 5-DOF onboard manipulator, letting operators command end-effector position for that pair instead of driving each joint separately.
- ***Tools & Techniques:*** *Object Detection, Real-Time Inference, Waypoint Navigation, Stanley Controller, Forward & Inverse Kinematics, YOLOv3.*

ACHIEVEMENTS

- Placed 2nd globally at the Anatolian Rover Challenge and won Best Autonomous and Controls (Jul 2022).
- Won the Excellence Award for Overall Performance at the International Rover Challenge, Bangalore (Jan 2023).